Single and Multilevel Inheritance - C++ Program

```
#include <iostream>
#include <string>
using namespace std;

// ========================================================
// 1. SINGLE INHERITANCE: Person (Base) -> Student (Derived)
// ========================================================
class Person {
protected:
    string name;
    int age;

public:
    void setPersonDetails(string n, int a) {
        name = n;
        age = a;
    }

    void displayPerson() const {
        cout << "Name: " << name << ", Age: " << age << "\n";
    }
};

class Student : public Person { // Single Inheritance
private:
    int rollNo;
    double marks;

public:
    void setStudentDetails(string n, int a, int r, double m) {
        setPersonDetails(n, a);
        rollNo = r;
        marks = m;
    }

    void displayStudent() const {
        cout << "--- Single Inheritance Output (Student) ---\n";
        displayPerson();
        cout << "Roll No: " << rollNo << ", Marks: " << marks << "\n\n";
    }
};
```

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Single and Multilevel Inheritance - C++ Program

```
// ========================================================================
// 2. MULTILEVEL INHERITANCE: Vehicle (Base) -> Car -> ElectricCar
// ========================================================================
class Vehicle {
protected:
    string brand;
    int maxSpeed;

public:
    void setVehicleInfo(string b, int speed) {
        brand = b;
        maxSpeed = speed;
    }

    void displayVehicle() const {
        cout << "Brand: " << brand << ", Top Speed: " << maxSpeed << " km/h\n";
    }
};

class Car : public Vehicle { // Level 1 Derived Class
protected:
    int numDoors;

public:
    void setCarInfo(string b, int speed, int doors) {
        setVehicleInfo(b, speed);
        numDoors = doors;
    }

    void displayCar() const {
        displayVehicle();
        cout << "Doors: " << numDoors << "\n";
    }
};
```

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Single and Multilevel Inheritance - C++ Program

```
class ElectricCar : public Car { // Level 2 Derived Class (Multilevel)
private:
    int batteryCapacity; // kWh
    int range; // km

public:
    void setElectricCarInfo(string b, int speed, int doors, int battery, int r) {
        setCarInfo(b, speed, doors);
        batteryCapacity = battery;
        range = r;
    }

    void displayElectricCar() const {
        cout << "--- Multilevel Inheritance Output (ElectricCar) ---\n";
        displayCar();
        cout << "Battery Capacity: " << batteryCapacity
             << " kWh, Range: " << range << " km\n";
    }
};

int main() {
    // Single Inheritance Demo
    Student s;
    s.setStudentDetails("John Doe", 20, 101, 88.5);
    s.displayStudent();

    // Multilevel Inheritance Demo
    ElectricCar eCar;
    eCar.setElectricCarInfo("Tesla Model S", 250, 4, 100, 600);
    eCar.displayElectricCar();
    return 0;
}
```

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